Exercise 01

What is the probability of measuring each of the following qubits in state |0> and state |1>?

$|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$, $|\phi\rangle = \frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$, $|x\rangle = \frac{1+i}{\sqrt{3}}|0\rangle - \frac{i}{\sqrt{3}}|1\rangle$

Exercise 02

Two vectors are given by:

$$|a\rangle = \begin{pmatrix} -2 \\ 4i \\ 1 \end{pmatrix} \qquad |b\rangle = \begin{pmatrix} 1 \\ 0 \\ i \end{pmatrix}$$

Find: $\langle a|$ and $\langle b|$ $\langle a|b\rangle$ and $\langle b|a\rangle$ $|c\rangle = |a\rangle + 2|b\rangle$

Exercise 03

Compute the norm of the following vectors: $|u\rangle = \begin{pmatrix} 2 \\ 4i \end{pmatrix}$, $|v\rangle = \begin{pmatrix} -1 \\ 3i \\ i \end{pmatrix}$

s.t: the norm is : $\|u\| = \sqrt{\langle u|u\rangle}$

Bra-ket and Quantum Gates: Practice Exercises

Conventions: $|0\rangle = (1, 0)^T$, $|1\rangle = (0, 1)^T$, $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$.

Part 1: Bra-ket basics

1. Write $|\psi\rangle = (1/\sqrt{2})(|0\rangle + i|1\rangle)$ as a column vector, then find the bra $\langle\psi|$ as a row vector.
2. Compute $\langle 0|1\rangle$, $\langle +|-\rangle$, and $\langle\psi|\psi\rangle$ for the state in problem 1.
3. Simplify $|0\rangle\langle 0| + |1\rangle\langle 1|$. What is this operator?
4. For $|\varphi\rangle = (\sqrt{3}/2)|0\rangle + (1/2)e^{i\pi/4}|1\rangle$, find the probability of measuring $|1\rangle$.

Part 2: Gates as outer products

5. Write X (NOT) and Z in outer-product form, using $|0\rangle\langle 1|$ and similar terms.
6. Compute $X|0\rangle$, $Z|+\rangle$, $H|1\rangle$, and $Y|0\rangle$, where $Y = -i|0\rangle\langle 1| + i|1\rangle\langle 0|$.
7. Show that HZH = X using matrices or the action on basis states.
8. Compute $S|+\rangle$, where $S = |0\rangle\langle 0| + i|1\rangle\langle 1|$. Which state on the equator of the Bloch sphere do you get?
9. Show that $H|0\rangle = |+\rangle$ and $H|+\rangle = |0\rangle$, and explain why $H^2 = I$.

Part 3: Two qubits

10. Compute $|0\rangle\otimes|1\rangle$ in the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.
11. Write CNOT (control first qubit) as $CNOT = |0\rangle\langle 0|\otimes I + |1\rangle\langle 1|\otimes X$.
12. Apply H on qubit 1, then CNOT, to $|00\rangle$. What state results? Is it entangled?
13. Apply CNOT to $|+\rangle|0\rangle$ and to $|+\rangle|-\rangle$. What does the second one show (phase kickback)?